STUDENT ARTICLE

STAKEOUT FROM SPACE: FOURTH AMENDMENT CONCERNS RESULTING FROM THE ONSET OF SATELLITE VIDEO

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I. INTRODUCTION

In 2014 a company called Skybox Imaging released video of a Beijing airport taken from Earth orbit using its satellite, the *Skysat-1*.[1] The resolution of the video was such that viewers were able to identify the exact flight number of the flights recorded landing.[2] Just months prior, the same company released video of a slightly lower spatial resolution[3] showing clips of the Earth's surface.[4] After amassing considerable investment revenue, and a name change and acquisition by Google[5], Terra Bella (formerly Skybox Imaging) was announced to be acquired by the larger Planet Labs (now known simply as "Planet"[6]), another operator of remote sensing satellites.[7] All the foregoing is to suggest that high definition video

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[1] Betsy Mason, *Incredible HD Video from Space Brings Maps to Life,* (Mar. 4, 2014), https://www.wired.com/2014/03/skybox-video-earth-from-space/.

[2] *Id.*

[3] Spatial resolution refers to the smallest object that can be discerned from the image. *See* http://www.nrcan.gc.ca/node/9407 for a more detailed explanation.

[4] David Wogan*, High-definition video from space is available for purchase. Finally.*, (Dec. 30, 2013), https://blogs.scientificamerican.com/plugged-in/high-definition-video-from-space-is-available-for-purchase-finally/.

[5] Google, *Google and Skybox Imaging Sign Acquisition Agreement*, (Jun 10, 2014), https://abc.xyz/investor/news/releases/2014/0609.html

[6] Tony Campitelli, *A Changing Planet: Meet Our New Brand*, (Jun 12, 2016), https://www.planet.com/pulse/meet-our-new-brand/ .

[7] Will Marshall, *Planet to Acquire Terra Bella from Google, Sign Multi-Year Data Contract*, (Fed. 3, 2017), https://www.planet.com/pulse/planet-to-acquire-terra-bella-from-google/.

from satellites in orbit of the earth is soon to be commercially available and the technology is being pursued with considerable material investment.

The privacy concerns of observing earth by satellite is a debate that has been discussed by scholars for decades,[8] but video surveillance by satellite is a capability that has existed within the realms of speculation regarding the government's ability and Hollywood dramatization.[9] Today's environment for commercial remote sensing services is distinctly different than the one envisioned by legal scholars decades ago.[10] We stand near in time to an era in which law enforcement (or a private citizen, perhaps) could conceivably purchase video from satellites capable of tracking the movements of an individual person. Certainly this presents legal issues different than those that may currently exist from the ability to view still images from satellite that are potentially years old- as is already possible with a service such as Google Maps.[11] This article intends to address some distinct concerns related to the Fourth Amendment that potentially arise with the use of video from space by analyzing hypothetical space video surveillance scenarios in light of older and more recent Supreme Court cases. In addition, this Article attempts to address the question of whether or not the commercial ubiquity of satellite video changes the debate of what constitutes a "reasonable expectation of privacy." Finally, this Article will propose that the Congress of the United States preemptively nullify much of the concern by legislating privacy rights as it relates to space video in a manner similar to the regulation of other law enforcement technologies.

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[8] *See generally* Krysten C. Kelly, *Warrantless Satellite Surveillance: Will Our 4th Amendment Privacy Rights Be Lost in Space,* 13 J. MARSHALL J. COMPUTER & INFO. L. 729, ii (1995), and Lisa J. Steele, *The View from on High: Satellite Remote Sensing Technology and the Fourth Amendment*, 6 HIGH TECH. L.J. 317, 334 (1991) (discussing the concerns of satellite remote sensing in the early Nineties).

[9] *See* BEHIND ENEMY LINES (20th Century Fox, 2001) (As part of the plot, the military uses live video feed from a satellite to track the movements of a downed Naval pilot).

[10] Steele, *supra* note 8, at 319-320.

[11] *See generally* https://www.google.com/maps (It is important to note that Google Maps and Google Earth are in fact two different services with similar satellite imagery).

II. EXAMINING SPACE VIDEO SURVEILLANCE UNDER RELATED SUPREME COURT CASES

a. Katz v. United States

Katz v. United States[12] is a 1967 Supreme Court case that expanded the definition of a "search" under the Fourth Amendment of the Constitution. In the case, a man was speaking on the phone to convey illegal wagering information while inside a public, but closed, phone booth.[13] The FBI placed a listening device on the exterior of the phone booth and was able to listen in on the conversation.[14] The defendant sought to argue that the phone booth was a constitutionally protected place and therefore, for the FBI to conduct the surveillance without a warrant was an unreasonable search. Arguably, then, the resulting recorded phone call should have been excluded from evidence under the exclusionary rule.[15] The FBI argued that because no physical intrusion had occurred, there could not have been a "search."[16] In writing the opinion of the Court, Justice Stewart claimed that both sides missed the mark in formulating the argument by focusing on the status of the phone booth,[17] and that, instead, what is protected from searches by the Fourth Amendment is what the person "seeks to preserve as private."[18]

From *Katz* we gather two important holdings: first, what may be searched depends somewhat on the subjective expectation of the person that the place or information will remain private, and second, that physical intrusion is not necessary to constitute a search, that is a search is entirely possible by remote, technological means. In his concurrence in *Katz,* Justice Harlan proffers a test for determining when a person has a reasonable expectation of privacy.[19] The test first requires that the person exhibited an actual and subjective expectation to withhold something as private, and secondly,

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[12] *Katz v. United States*, 389 U.S. 347 (1967).
[13] *Id* at 348.
[14] *Id*.
[15] *Id* at 350.
[16] *Id* at 352.
[17] *Id* at 351.
[18] *Id* at 352.
[19] *Id* at 361.

that this is an expectation that "society is prepared to recognize as reasonable."[20]

Although it cannot be done, due to later Fourth Amendment cases related specifically to aerial surveillance (which will be discussed later in the Article), it is helpful to examine video surveillance from space as if the *Katz* holding existed in a vacuum. Hypothetically, assume someone were to have a house on a busy public street in the center of Manhattan. Assume further that they were to do some activity by an open window of that home. If a government agency were to record that activity from the street outside one could argue that it would be admissible evidence because, while that person may have satisfied the first prong of the reasonable expectation of privacy test, the court would likely hold that society would not find that expectation as reasonable. Now imagine someone who lives in a house in a small acre clearing of a forest (the forest also being owned by the person), far from any public road, and with the entirety of that clearing obscured from the road. That person could reasonably expect that his or her actions within the home or clearing are private, and society would most likely find that expectation reasonable. Which would conceivably mean for any law enforcement to observe anything in the home or the clearing would require a warrant to enter and search.

When you add in the prospect of aerial surveillance, by satellite or otherwise, this hypothetical becomes considerably more difficult to analyze. May a person reasonably expect privacy when airplanes overhead could photograph or record his or her activities in the clearing? If so, would society deem that expectation reasonable? Later Supreme Court cases have held that, society does not find such expectations to be reasonable.[21] In general, if law enforcement uses aerial (by plane or helicopter) surveillance, even without a warrant, it is not a violation of a person's Fourth Amendment rights.[22] These forms of aerial surveillance are reasonably detectable by the one under surveillance, at least to some extent. Would a

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[20] *Id*. (This test was later adopted by the Supreme Court in *Smith v. Maryland*, 442 U.S. 735, at 740.)

[21] *See generally Florida v. Riley,* 488 U.S. 445 (1989) and *California v. Ciraolo*, 476 U.S. 207 (1986) (both cases involving aerial surveillance) and *Dow Chemical v. United States*, 106 S. Ct. 1819 (1986) (Also involving aerial surveillance, although somewhat limited to a commercial context.)

[22] *Id*.

pure *Katz* analysis change when the method of surveillance on our hypothetical house in the clearing is silent and undetectable, as would be the case with satellite video? One may instinctively begin to ask questions about the frequency and altitude of the flights over the clearing. If there are any flights at all, one may question whether society would reasonably expect privacy in the clearing - perhaps certainly in the walls of the home, but less reasonable in the open of the clearing. If the method of surveillance is silent and effectively invisible, it is hard to expect any rational person (devoid of any other knowledge) to assume that someone might witness their actions in the clearing. Suppose that a satellite video service exists in a manner similar to and nearly as ubiquitous as Google Maps; will the Supreme Court then interpret society as generally being aware that satellite video surveillance is available, and therefore society *would not* hold an expectation of privacy in the clearing reasonable? As one can see, *Katz* conceivably raises more questions than answers for the future law enforcement application of space video.

Cases after *Katz,* such as *Oliver v. United States*, seemingly affirm this return to the pre-*Katz* distinction of *where* the location the "search" occurred being the defining factor in whether a constitutional search has actually occurred.[23] Although, the same case appears to reserve some constitutional respect for the privacy of one's "curtilage," that is, the land directly adjacent, or considered part of, the home.[24]

b. Aerial Surveillance in Ciraolo and Riley

The largest blows to those wanting protection from law enforcement's eventual use of satellite video may have already been dealt. The summation of the cases *California v. Ciraolo, Florida v. Riley,* and *Dow Chemical v. United States* give rise to the "Plain View Doctrine."[25] Although *Oliver* seemed to protect the curtilage

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[23] *See Oliver v. United States,* 466 U.S. 170, at 179 (1984) (holding that "Only the curtilage, not the neighboring open fields warrants the Fourth Amendment protections that attach to the home.")

[24] *Id* at 180.

[25] Kelly, *supra* note 8, at 754 ii.

of one's home, these three cases apparently destroy that protection.[26] In *Ciraolo*, a defendant was convicted of growing marijuana in his yard after local law enforcement received an anonymous tip, flew an airplane overhead at 1000 feet, and photographed the marijuana plants without a warrant. [27] In the majority opinion Justice Burger writes:

> That the area is within the curtilage does not itself bar all police observation … The observations by Officers Shutz and Rodriguez in this case took place within public navigable airspace … in a physically nonintrusive manner; from this point they were able to observe plants readily discernible to the naked eye as marijuana.[28]

In what many now refer to as the "naked-eye" test, *Ciraolo* has removed the curtilage of one's home from the undeniable need for law enforcement to obtain a warrant to search. In effect, *Ciraolo* deemed the "public navigable airspace" to be a location from which a person cannot expect privacy, even within his curtilage.[29] Which begs the question: is the orbit of our earth a proper extension of "public navigable airspace"? If yes, the result does not bode well for those wanting warrants in order for law enforcement to utilize video gathered from satellites. Some authors are inclined to believe that this warrantless use of satellite imagery will be the norm, because the altitude at which the flight occurs seems unimportant if the technology required to photograph in detail at that altitude is irrelevant.[30]

As an additional note, the California Court of Appeal (from which the case was granted *certiorari*) noted that the flight from which the plants were photographed was undertaken specifically for the purpose of surveilling the defendant's particular home and was distinct from a civilian flight overhead, which would have merely happened to observe the plants.[31] The Supreme Court's

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[26] *Id* at 753. *See also*: Daniel W. Johnson, *Aerial Surveillance: A Birds-Eye View of Katz v. United States*, 22 Gonz. L. Rev. 393, at (1986)

[27] *Ciraolo*, 476 U.S. at 209.

[28] *Id*. at 213.

[29] *Id.*

[30] Kelly, *supra* note 8, at 753 (Quoting *Dow* in which the Court held that "the mere fact that vision is enhanced somewhat… does not give rise to constitutional problems).

[31] *Ciraolo,* 476 U.S. at 211.

holding renders this distinction irrelevant.[32] *Florida v. Riley* reaffirmed this holding three years later when a person growing marijuana in their greenhouse had that marijuana photographed by law enforcement in a helicopter at an altitude of 400 feet.[33]

c. Does the Technology used Matter? – Dow Chemical

Dow Chemical v. United States sheds uncertainty on the "naked-eye" test we derive from *Ciraolo* and *Riley*. In *Dow Chemical*, the EPA used an airplane to observe and perform an unconsented inspection of a Dow Chemical plant in Midland Michigan.[34] The record states that the aircraft contained a "standard floor-mounted, precision aerial mapping camera."[35] *Dow Chemical* is invaluable to the present analysis because it offers considerable discussion to the type of technology used. *Ciraolo* was clear: the surveillance of defendant's backyard was constitutional because it was done from public airspace and involved material visible by "the naked-eye."[36] This stands in stark contrast to *Dow Chemical* where the "search" was conducted with a "precision aerial mapping camera" capable of discerning objects a half inch in size.[37] This discrepancy between the two holdings is seemingly only mendable by concluding that the Court intends advanced technology to stand as the "naked-eye." Consequently, there does not seem to be a situation in which the technology used to surveil would ever matter.[38] So, it, perhaps unfortunately, follows that use of satellite video could easily become an extension of "navigable airspace" and the highly refined, expensive telescopic video cameras onboard these satellites are, inexplicably, an extension of the "naked eye."

The *Dow Chemical* opinion mentions that the Government concedes that there could be a cut-off in terms of technology (offering satellites as an example) that would be considered outside the

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[32] *Id.* at 215.
[33] *Riley*, 488 U.S. at 446.
[34] *Dow Chemical*, 476 U.S. at 229.
[35] *Id.*
[36] *Ciraolo*, 476 U.S. at 209.
[37] *Dow Chemical*, 476 U.S. at 238.
[38] Daniel W. Johnson, *Aerial Surveillance: A Birds-Eye View of Katz v. United States*, 22 Gonz. L. Rev. 393 (1986), at 403.

"naked-eye" and thereby require a warrant.[39] Although the Supreme Court is merely restating the government's argument, it does seem interesting that it would even mention the Government's reference. It is almost as if the Court does not wish to outright hold that all forms of enhanced vision will be considered the "naked-eye" and would like to reserve some maneuvering room for the future. The *Dow Chemical* opinion does, however, seem to take the general availability to the public of the surveillance method into consideration when determining if an unreasonable search occurred.[40] As some commenters have noted, it is almost incredulous the court would find a nearly "$22,000 aerial mapping camera" generally available to the public, as if "any Joe Sixpack with an airplane and [the camera] could readily duplicate these photos."[41] Nevertheless, such an interpretation invites the question of whether or not the same court would hold a satellite to be a similar piece of "generally available" technology. If satellite video were to exist for some time as a relatively unknown (to the general public) technology, then perhaps the same court would find its warrantless use by law enforcement an unreasonable search. But if satellite video were to become as ubiquitous as Google Maps, as this author imagines it might, then would the availability of the technology erode the protection the Fourth Amendment offers against it? So, while this Article intends to focus on law enforcement's potential use of satellite video, *Dow Chemical* seems to suggest that the general public's use of the technology involved plays a role in determining whether or not a Fourth Amendment search has occurred.

d. Using Satellite Video to Invade the Home – analogous to Kyllo?

Although satellites obviously exist outside of homes and would provide video from an overhead view, the fact that it is video rather than still imagery may allow one to deduce the contents inside of the house. Consider our earlier home in the clearing hypothetical. If law enforcement wished to surveil the owner using Google Maps, they would not be able to determine much. If something illegal was

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[39] *Dow Chemical*, 476 U.S. at 238.
[40] *Dow Chemical*, 476 U.S. at 234.
[41] Johnson, 22 GONZ. L. REV. 393, at 403 (quoting Professor Yale Kamisar.)

conspicuously located in the clearing the moment the satellite took the still image, then perhaps law enforcement would have a case to investigate or prosecute. But even then, the still image may be old and therefore inconclusive. On the other hand, if the law enforcement had video at its disposal it could possibly record a vehicle leaving every day at 8:00 a.m. that does not return until 5:00 p.m., thereby allowing them to deduce that in between those two times, at least one person *is not* in the home. Or, suppose satellite video captures a van arriving at the home every day at a certain time and captures the driver carrying some object from the van into the home. If law enforcement uses the video to track the van as it goes to and from a known marijuana growing operation, then they can deduce with reasonable effectiveness that there is marijuana inside the home. Again, we see how satellite video is distinct from other forms of satellite surveillance and requires its own consideration.

In *Kyllo v. United States*, law enforcement suspected a man of growing marijuana within his house.[42] Knowing that high powered lamps are often used to grow marijuana indoors, the law enforcement used (without a warrant) a thermal imaging camera to see if there were unusual heat concentrations emanating from the house.[43] There were heat concentrations, and law enforcement subsequently were issued a warrant to enter the home where they found marijuana growing.[44] The defendant in the case was arrested and sought to have the evidence suppressed because the thermal scan of the home was an unreasonable search under the Fourth Amendment.[45] The syllabus from the case gives us the clearest interpretation of the holding:

> Where, as here, the Government uses a device that is not in general public use, to explore details of a private home that would previously have been unknowable without physical intrusion, the surveillance is a Fourth Amendment 'search,' and is presumptively unreasonable without a warrant.[46]

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[42] *Kyllo v. United States*, 533 U.S. 27, at 29 (2001).
[43] *Id.*
[44] *Id.* at 30.
[45] *Id.*
[46] *Id* at 29.

Such a holding applied to satellite video seems workable in its favor. Satellite video is certainly a device that is not currently in general public use, and in a situation like our home in the clearing hypothetical, it could certainly be used to deduce the details of the home in the clearing; details which would otherwise be unknowable without physically entering the property. In addition, this holding seems to support the holding of *Dow Chemical* in that the reasonableness of the expectation of privacy depends, at least to some extent, on the general availability of the technology used to conduct the surveillance.[47] One of the criticisms of those in the dissent was that the majority provided no guidelines as to what constitutes "general use."[48] This gives citizens and law enforcement no guidance on how video gathered from satellites may be utilized. At what point will satellite video cross the line into "general use" of the public, and thereby society find the expectation that activities in the clearing is unreasonable? I argue that point will come whenever the average citizen can go to a service and order video from a specific time and specific location; especially given that surveillance requiring a $22,000 camera and an airplane was once held to be "generally available to the public."[49]

Even if satellite video does become generally available to the public and useful to make deductions about the contents inside the home, should society still legislatively proscribe its warrantless use? The opinion of *Kyllo* suggests this may not be necessary, as Fourth Amendment jurisprudence would suffice. The *Kyllo* opinion states that the Fourth Amendment "draws a firm line at the entrance to the house."[50] The obvious counter to this would be that police evidence has long been gathered by staking out homes and deducing their contents by the traffic that comes and goes from them.[51] If the positioning of the video satellite is considered a public area, "navigable airspace," remaining consistent with *Ciraolo* and

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[47] Thomas D. Colbridge, *Kyllo v. United States: Technology versus Individual Privacy*, 70 FBI L. Enforcement Bull. 25, 28 (2001).

[48] *Id.*

[49] Johnson, 22 GONZ. L. REV. 393, at 403.

[50] *Kyllo*, 533 U.S., at 40 (quoting *Payton,* 445 U.S., at 590).

[51] *See Horton v. California*, 496 U.S. 128, at 137 (1990) (the court upheld warrantless seizure when "…the officer [is] *lawfully located* in a place from which the object can be plainly seen…" demonstrating an emphasis on searches being lawful without warrant when conducted by an officer who is located in a place where he has a lawful right to be).

Riley, and it's "generally available to the public," remaining consistent with *Dow* and *Kyllo*, then it's hard to imagine how our federal courts *would not* hold the use of satellite video to be any different from a police vehicle staking out a home. The distinction in *Kyllo* seems to be that using a generally *unavailable* technology to determine what is *inside* the home constitutes a warrant-requiring search. This author's envisioned use of satellite video (prior to its general availability, at least) fits that bill rather nicely. Yet in light of *Ciraolo*, one can just as easily envision the warrantless use of satellite video being upheld as not a search. This conundrum may lead to considerable unpredictability for law enforcement when the day comes that they are in a position to utilize satellite video when investigating a suspect.

Further indicative of the mess that is Fourth Amendment jurisprudence is the fact that *Kyllo* was heard by the Ninth Circuit Court of Appeals four times, decided on the third, and then the Ninth Circuit reversed itself on the fourth hearing, holding that the search *was* constitutional.[52] It's difficult even for the highest levels of our federal courts to come to a consensus as to what factors make a search unreasonable.

e. Using Satellite Video to Track Movements – analogous to Jones?

One of the additional uses the reader must envision satellite video providing is the tracking of individuals. Suppose law enforcement had sufficient information to suspect a person of committing a crime, but not enough to give them probable cause or a warrant. So, they set out to track the person's vehicle, on the public roadways, where generally one does not have a reasonable expectation of privacy,[53] all in the hope of gathering information about the suspect's activities. Satellite video, even in its current form (as detailed in the introduction of this Article), is capable of distinguishing individual vehicles. If there were enough satellites so that they could "hand-off" the video when one passes out of range, and the other into range, then law enforcement could conceivably track an individual's car silently, undetectably, and without needing to touch the

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[52] *Id.*

[53] *See generally, United States v. Knotts*, 460 U.S. 276 (1983).

car to place a tracking device or recover or intercept its on-board location data. This ability could be particularly useful to law enforcement, and in some instances where other methods of aerial surveillance are not feasible or the law enforcement has a particular need to remain undetected, satellite video could even be chosen over airplane, helicopter, or drone surveillance.

In 2012, the Supreme Court decided *United States v. Jones*.[54] *Jones* involved law enforcement tracking a suspect via a Global Positioning System (GPS) device attached to his car for a four week period.[55] The defendant sought to have the evidence gathered by the GPS tracking suppressed, and initially the District Court would only suppress the data from when the car was parked at the defendant's home.[56] For all the other data, the Court claimed while on public thoroughfares the defendant had no reasonable expectation of privacy.[57] Ultimately, the case made its way to the Supreme Court where the Court held that the attachment of the device to the car constituted a search for the purposes of the Fourth Amendment.[58]

Interestingly, the case was not decided on a reasonable expectation of privacy grounds, but instead, on the theory that placing the GPS tracker on the car constituted a trespass to chattel and therefore a search.[59] Justice Scalia in the majority opinion, specifically mentions *Katz* and decides to sidestep its "reasonable expectation of privacy" test, instead choosing to return to pre-*Katz* jurisprudence related to trespass.[60] To apply satellite video directly to *Jones* leads one to few conclusions. Because *Jones* was decided solely on the issue of trespass, it does not answer whether or not constant and long-term 24/7 tracking is actually unconstitutional without a warrant.[61] And, because satellite video surveillance certainly would not involve a physical trespass, the jury is still out, so

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[54] 565 U.S. 400 (2012).
[55] *Id.* at 403.
[56] *Id.*
[57] *Id.*
[58] *Id.* at 402.
[59] *Id.* at 404.
[60] *Id.* at 406-7.
[61] *See,* Heather Phillips*, The Big Brother Effect: The Implications of the Unanswered Question in United States v. Jones,* 48 MCGEORGE L. REV. 395, 422 (2017) (Noting that the majority opinion in *Jones* did not address at what point warrantless police surveillance or tracking becomes a warrant-requiring search).

to say, on whether or not satellite video tracking would be *per se* unconstitutional without a warrant.

Those who may be worried about a future of satellite video in the hands of law enforcement may find relief in the concurrences of *Jones*. The Jones case spawned two concurrences, each generally more favorable to addressing long term individual tracking by means of a post-*Katz* reasonable expectation of privacy test, rather than simply traditional trespass means.[62] If satellite video ultimately is capable of tracking the individual, physical trespass doctrines will do nothing to stop it. Justice Sotomayor is distinctly aware of this fact, noting in her concurrence that other forms of individual tracking will not require physical trespass at all.[63] Justice Alito also considers that non-physical trespass (even assisted by aerial surveillance) is afforded no protection at all by the majority in *Jones*.[64] Justice Alito, in his concurrence, which four other justices joined, wishes to do away entirely with pre-*Katz* physical trespass as a test of when one's right to privacy has been abridged, instead relying on a reasonable expectation of privacy test.[65] Sotomayor, alone in her concurrence, would still like to recognize physical trespass as constituting a search, but thinks the pervasive individual tracking should have been subject to the reasonable expectation of privacy test.[66] The issue for her, is that when the government is able to collect a "comprehensive record of a person's public movements that reflects a wealth of detail about her familial, political, professional, religious, and sexual associations" it is ripe for abuse.[67] Further, it could change the way the government and its citizens interact: "Awareness that the Government may be

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[62] *See generally* David Gray; Danielle Keats Citron; Liz Clark Rinehart, *Fighting Cybercrime after United States v. Jones*, 103 J. CRIM. L. & CRIMINOLOGY 745, 762-764 (2013) (noting that both Justic Sotomayor's and Justice Alito's concurrences are both sympathetic to modifying 4th Amendment privacy doctrine in the digital age.)

[63] *U.S. v. Jones*, 565 U.S. at 414.

[64] *Id* at 425.

[65] *See Id* at 422-23 (Alito notes that *Katz* "did away with" the old doctrine, and notes several cases pointing to the abandonment of physical trespass-based tests for unreasonable searches; quotes *United States v. Karo*, 468 U.S. 705, 713 (1984) in which the court held "an actual trespass is neither necessary nor sufficient to establish a constitutional trespass").

[66] *Id* at 414.

[67] Gray et al., supra note 62, at 763.

watching chills associational and expressive freedoms."[68] Some of Justice Sotomayor's concurrence might suggest a willingness to even abandon the public-observation doctrine.[69] Justice Alito also questions the public view doctrine's survival in the modern age of law enforcement surveillance.[70] Abandoning this doctrine would be beneficial to protecting against unreasonable searches conducted via satellite. It would allow for a reasonable expectation of privacy to exist against some forms of surveillance, even if the person is acting in public view, if the surveillance and individual tracking is constant and ongoing. If satellite video surveillance can be conducted constantly and conducted of a person in public view, then there is little to suggest that their Fourth Amendment right has been violated. But, if the court adopts a mosaic theory of privacy, as Justices Sotomayor and Alito suggest,[71] then perhaps in the future Fourth Amendment jurisprudence will be more favorable to satellite video surveillance protections.

Although this is not position of the Supreme Court at the time, the fact that both concurrences address the issue of constant tracking and surveillance being the "unreasonable search" seems to suggest that, going forward, the Court is in position to accept that constant surveillance can be a search even done in public areas.

Regardless, this author reasons that GPS tracking and potential satellite video tracking are not perfect analogues. Satellite video would suffer from numerous limitations: inability to penetrate cloud cover,[72] losing vehicles in tunnels or other covered areas, gaps in swath coverage,[73] overall lack of satellites, etc. Part of the reasoning behind the *Jones* decision (at least by the concurring Jus-

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[68] *US v. Jones,* 565 U.S. at 416.

[69] *Id*.

[70] *Id* at 764.

[71] *Id* at 762.

[72] *See* NASA Earth Observatory, https://earthobservatory.nasa.gov/IOTD/view.php?id=79233 (last visited April 9, 2017) (Demonstrating that when imaging in the visible spectrum, NASA's EO-1 satellite cannot image the ground through smoke; imaging through it requires using more than the visible spectrum).

[73] *See* Allan Brimicombe*, GIS, Environmental Modeling and Engineering Second Edition*, at 40 (CRC Press, 2009) (Describes generally what swath width is and, as an example, describes how the IKONOS satellite would require multiple passes over London in order to image its entirety).

tices) was the completeness, almost omnipotent nature of the tracking that the GPS provided.[74] In that sense, satellite video would most likely be lacking, relative to GPS. It is easy to imagine law enforcement knowing the unlawful nature of warrantless GPS tracking (if the police placed a tracker on the car, that is) and thus seek satellite video tracking as an alternative. It is just as easy to imagine a day in the future when, by technological advancement, satellite video has largely overcome some of the deficiencies posited above; bringing it just as, or nearly as, effective at tracking an individual's movements as GPS. This is simply another reason to suggest that legislative bodies address the situation rather than wait for the higher levels of our federal and state courts to inevitably weigh in on the matter.

III. LEGISLATING LAW ENFORCEMENT'S USE OF SATELLITE VIDEO

As one can see from examination of the relevant Fourth Amendment Supreme Court Cases, we are left with little certainty as to how the law will eventually apply law enforcement's use of satellite video to Fourth Amendment cases. This has been the case with other technologies, and, in response, state legislatures and the federal legislature have enacted legislation and regulation governing the use of certain technologies in relation to their use by law enforcement.[75] This author suggests the same should be done with satellite video. In order to examine how this might take place, this section of the Article will examine some technological analogues and how their use has been addressed in legislation.

There is debate as to whether or not legislatures should regulate what would normally be Fourth Amendment protections at all.[76] On one side (and the side to which this author is partial) schol-

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[74] *See Jones*, 565 U.S. at 415 (Justice Sotomayor, in concurring noting the litany of personal information that be gathered about someone from the constant, long term tracking of their vehicle) *and also, Id.* at 430 (Justice Alito, in his concurrence noting that "longer term GPS monitoring in investigations of most offenses impinges on expectations of privacy").

[75] David E. Steinberg, *Sense-Enhanced Searches and the Irrelevance of the Fourth Amendment*, 16 WM. & MARY BILL RTS. J. 465, 472 (2007)

[76] *Id* at 473.

ars disfavor the unpredictability of the Fourth Amendment jurisprudence.[77] As Fourth Amendment scholar David Steinberg notes, there is an inability to predict how the Supreme Court will address a new technology; prior to the handing down of the *Kyllo* decision he predicted the court would not require a warrant, but ultimately the court did.[78] Steinberg asks the question that if he, who has spent years studying the Fourth Amendment, cannot predict how the Court will address a technology "then why should we expect police officers to make accurate predictions?"[79] Therein lies the crux of this author's argument: in leaving satellite video to the Supreme Court, we make the everyday law enforcement officer the "judge" as to what is a reasonable search, at least until the courts have ruled conclusively on the new technology. To leave that determination in such hands is to subject it to decision-making with considerable bias. Further, the judicial system is inherently slow, and by the time the satellite video evidence reaches the bench, the searched person's right may have already been violated.

Conversely, some scholars suggest that new technology under the Fourth Amendment should be left to the courts.[80] Steinberg points out that those in this camp tend to disfavor the complexity that arises from legislative actions. Instead, they prefer that the basic Fourth Amendment principles are relatively accessible and concepts such as "reasonable expectation of privacy, probable cause, and reasonable suspicion are easily understood.[81] This author would argue that those with this position are possibly overestimating the average person's grasp of these concepts. In other words, they *can* be easily understood by the average person, but are they *actually* understood by the average person?

If that supposition is correct, that the average person may not truly understand what a reasonable expectation of privacy is (along with other Fourth Amendment concepts), then it makes it all the more important that we find a legislative solution to law enforcement's use of satellite video. At the Federal level, wiretapping was

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[77] *Id.*
[78] *Id.*
[79] *Id.*
[80] *Id* at 475.
[81] *Id.*

regulated in the aftermath of *Katz*.[82] Wiretapping gives the law enforcement the ability to listen or record the content of a phone call. Congress passed the Omnibus Crime Control and Safe Streets Act of 1968 which dictated that law enforcement's use of wiretaps would require a warrant.[83] In 1986, Congress enacted roughly the same requirements for pen registers (devices that capture to whom, and from where a phone call was made).[84]

Drones are a more relevant and recent example of a technology undergoing some legislation and regulation. At the Federal level there has been a push to regulate law enforcement's use of drones, with a few failed acts having warrant provisions.[85] At the state level, there has been some legislation successfully passed to require warrants for their use by law enforcement. In 2013, Illinois, Florida, Idaho, Montana, Oregon, Tennessee, and Texas all passed drone legislation that included some form of warrant-requiring provision.[86] The requirement is not always absolute; the states' legislation provides exceptions for emergencies, "exigent circumstances," and to forestall the escape of a suspect, among other exceptions.[87] There is a trend among the states to allow the legislature to more clearly define, for citizens, what exactly constitutes their reasonable expectation of privacy in relation to drones.

This author proposes that Congress follows the lead of states' action related to drones and define for the entirety of the United States what constitutes a reasonable expectation of privacy in relation to law enforcement's use of satellite video. In fact if you were to take 18 U.S.C. § 2518 (the warrant requirements for wiretapping) and replace every instance of "the interception of a wire, oral, or electronic communication" with "surveillance of an individual or individuals by means of recorded video from an orbiting satellite" you would be very near to an acceptable warrant-requiring provision for satellite video. Whatever Congress decides, it is best to do

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[82] *Id* at 471.

[83] *Id.* (for the current version of the law see 18 U.S.C. § 2518).

[84] *Id* at 472.

[85] Thomas A. Bryan, *State v. Brossart: Adapting the Fourth Amendment for a Future with Drones*, 63 CATH. U. L. REV. 465, at 494 (2014).

[86] Taly Matiteyahu, *Drone Regulations and Fourth Amendment Rights: The Interaction of State Drone Statutes and the Reasonable Expectation of Privacy*, 48 COLUM. J.L. & SOC. PROBS. 265, 285 n.145 (2015).

[87] *Id* at 285-286.

so before the technology matures in order to prevent the infringement of constitutional protections prior to the issue being dealt with by the courts.

IV. CONCLUSION

The concerns addressed in this Article may be more distant than they are near. Regardless, satellite video is a currently existing technology (even if only in its infancy), the further development of which is being pursued with considerable investment. Once it is developed to maturity, it could serve as a useful tool for law enforcement to conduct searches, track individuals and groups, and otherwise conduct investigations. Like any technology, it is a tool at law enforcement's disposal. However, law enforcement often will rush in to using the new technology without sufficient regard for Fourth Amendment principles. Perhaps rightfully so, law enforcement's mission is generally to pursue criminal suspects and protect the public, not to debate the minutiae of how each and every technology fits into Fourth Amendment jurisprudence.

Examining how satellite video surveillance fits into Fourth Amendment concepts and jurisprudence would not only serve the public's interest in being protected against unreasonable searches, but also benefit law enforcement by clearly defining their role and minimizing losses of evidence due to the exclusionary rule. When one attempts this, however, they are left with more questions than answers. *Katz* gave birth to the reasonable expectation of privacy test as well as confirming that an unreasonable search can be conducted via remote, technological means, seemingly suggesting that the fact that a satellite video is taken remotely does not preclude protection against its warrantless use. Just after arriving at that conclusion, we are faced with *Ciraolo* and *Riley*, where the warrantless use of overhead aerial surveillance was held to *not* infringe upon a constitutional protection. This leads one to inevitably ask whether or not our courts will find satellite video to be merely another form of aerial surveillance, making warrants for its use unnecessary. *Dow Chemical* expanded upon *Ciraolo* and *Riley* by finding that the use of highly specialized camera equipment to conduct the aerial surveillance did not it any way change the way the court applied the reasonable expectation of privacy test. This seemingly

allows for not only warrantless aerial surveillance, but aerial surveillance with highly specialized, sense-enhancing equipment.

Kyllo appears to offer some protection against warrantless satellite video surveillance by holding that it is an unreasonable search to use technology that is not in general public use to discern the private details of the home without a warrant. As the dissent in that case pointed out, there are no guidelines as to what constitutes general public use. How would the court determine when satellite video has reached the level of "general public use?" Further, that holding seems limited to when the technology is used to discern what is *inside* the home. Does using satellite video to watch persons coming and going from a home, and thereby deduce the contents of the home count as invading the home? Finally, in *Jones* the court held constant and long term warrantless tracking of an individual to be an unreasonable search, but only due to the physical trespass required to place the tracking device on the person's car. Does that mean if satellite video can achieve the same result (in terms of tracking) without requiring a physical trespass that its warrantless use would be upheld under the *Jones* holding?

The courts have struggled to adapt the Fourth Amendment as technology continues to evolve and surveillance becomes easier to conduct and more pervasive. This could simply be a result of the drafters of the Fourth Amendment never intending or conceiving the technology that is available today. A definitive, legislative solution can neutralize some of the worry satellite video may present. Federal and state legislatures have enacted legislation to specifically require warrants when law enforcement wishes to use pen registers, wiretaps, drones, and other technologies. There is little reason to avoid doing the same for satellite video.